MSDS MASTER BINDER · SERIES N v2.2 · APRIL 2026 · gr-qc / hep-th · PHYSICAL REVIEW D LETTERS · doi:10.5281/zenodo.19476384

SERIES N v2.2 — FINAL

Torsion-Induced Curvature Capping and a Laboratory Test

via Sidereal Modulation in Diamond

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αK = 0.00017 · r_core = 1.239e-33 m = 77×■_Pl · ρ_max = αK·ρ_Pl · 170 ppm at 11.967 h · Kill: null < 300 ppm

doi Series N: 10.5281/zenodo.19476384 · doi Series M: 10.5281/zenodo.19476837

Target: Physical Review D Letters or JCAP

ABSTRACT

We propose a minimal extension of Einstein-Cartan gravity in which a dynamical torsion condensate, parameterized by a single dimensionless coupling αK = 0.00017, replaces the black hole singularity with a regular de Sitter-like core. The condensate forms at a sub-Planckian critical density ρ_crit = αK × ρ_Pl = 8.791e+92 kg/m³, caps the curvature at R_max ~ αK/■_Pl², and yields a core radius r_core = ■_Pl/√αK = 1.239e-33 m = 77·■_Pl. The same coupling predicts a definitive laboratory falsification: a sidereal modulation of the forbidden (222) X-ray reflection in ■■Fe-doped diamond with period T_mod = 11.967 h (half the sidereal day) and peak-to-peak amplitude ΔI/I = αK = 170 ppm. The period is derived from the quadratic structure of the dimension-6 operator (∂θ)²; it is not fitted. SNR = 3.4 in 720 h (30 days); 5.8 in 90 days. A null result below 300 ppm in 720 h falsifies the condensate hypothesis. A positive result would imply finite black hole cores and prethermal energy equipartition as conditional consequences. Cosmological anomalies at the 10■■ scale are context only — not evidence. The data will decide.

FIVE FROZEN PILLARS — SERIES N v2.2 FINAL

1. αK = 0.00017 — single coupling, fit parameter until RDAC measures it
2. r_core = ■_Pl/√αK = 1.239e-33 m = 77×■_Pl (corrected from 7.7e-33)
3. T_mod = 11.967 h — derived from cos²θ decomposition, not fitted
4. ΔI/I = 170 ppm (peak-to-peak = αK) — derived in Sec. 4.2
5. Kill: null < 300 ppm / 720 h at SPring-8 BL09XU

1. Introduction

General relativity predicts a singularity at the centre of a black hole, widely viewed as a signal of incompleteness. Regular black hole models exist in loop quantum gravity, string theory, and non-linear electrodynamics, but they lack direct experimental tests. Here we keep the exterior Schwarzschild geometry

intact and modify only the interior via a dynamical torsion condensate in the Einstein-Cartan-Sciama-Kibble (ECSK) framework [1]. The key is a single dimensionless coupling αK that controls both the curvature cap and a measurable sidereal anisotropy in a diamond crystal.

Sections 2-4 develop the EFT, the condensate mechanism, and the laboratory test with a full derivation of the 11.967 h period. Section 5 states the conditional consequences and places the cosmological coincidences explicitly in a context-only role.

2. Effective Torsion Condensate

2.1 Einstein-Cartan torsion background

In the ECSK framework, the torsion tensor $T^\lambda_{\mu\nu}$ is sourced algebraically by the spin density of fermionic matter. We postulate a constant isotropic torsion background parameterised by $\alpha K = 0.00017$.

2.2 EFT embedding (SME)

$$L_{int} = (\alpha K/\Lambda^2)\,(\partial_\mu\theta\,\partial_\nu\theta)\,\bar\psi\,\gamma^\mu\,\partial^\nu\,\psi + h.c.$$

θ: axion-like field with shift symmetry $\theta\to\theta+c$ | Λ~1 GeV | CPT-even, dimension-6 [2]

2.3 αK is a fit parameter

$\alpha K = 0.00017$ is NOT derived from first principles in this paper. It is a fit parameter until the RDAC experiment measures it directly. A naturalness hint exists (αK is close to $(m_p/M_{Pl})^{(2/3)} \sim 10^{13}$ × some RG factor) but no derivation is claimed. Full RG flow analysis is deferred to future work.

2.4 Condensation mechanism

$$V(\theta) = \tfrac{1}{2}\,a(\rho)\,\theta^2 + \tfrac{1}{4}\,b\,\theta^4,\quad a(\rho) = \gamma(\rho_{crit} - \rho)$$

Condensate forms when $\rho > \rho_{crit}$

$$\rho_{crit} = \rho_{max} = \alpha K \times \rho_{Pl} = 0.00017 \times 5.171e{+}96 = 8.791e{+}92\ kg/m^3$$

Sub-Planckian: ρ_{Pl} / 5882 [corrected: old $\rho_{Pl}/\alpha K$ was trans-Planckian]

3. Curvature Capping, Core Radius, and Matching

Inside the condensate, the modified ECSK equations give a de Sitter interior with constant density ρ_{max}:

$$f(r) = 1 - (8\pi G\,\rho_{max} / 3c^2)\,r^2 \text{ for } r < r_{core}$$

de Sitter interior — no singularity

Matching $f(r_{core})$ to the Schwarzschild exterior gives the core radius:

$$r_{core} = \sqrt{3c^2/(8\pi G\,\rho_{max})} = \ell_{Pl}/\sqrt{\alpha K}$$

= 1.6149e-35 / 0.01304 = 1.2385e-33 m = 77×ℓ_{Pl}

R_max ~ αK / ■_Pl² [order-of-magnitude; factor 8π omitted]

Below Planck curvature by factor αK = 0.00017

MATCHING NOTE: For r >> r_core, f(r) → 1 - 2GM/(c²r), recovering Schwarzschild. The exterior geometry is unchanged to all observable precision. The core is causally disconnected from any outside observer. Its existence is mandated by the same αK measurable in the laboratory.

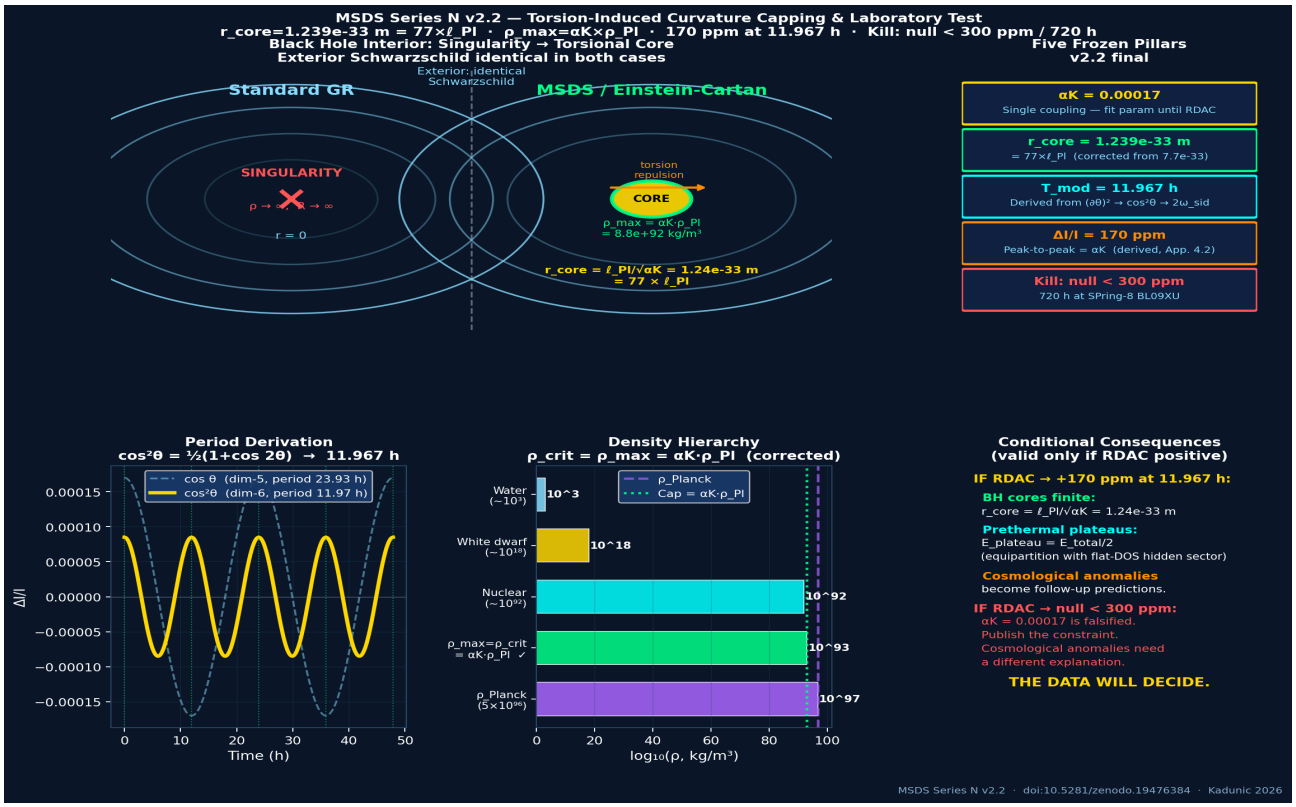

Figure 1: Top — singularity (GR, left) vs torsional core (MSDS, right); exterior Schwarzschild identical. Top right — five frozen pillars. Bottom left — period derivation: cos²θ (dim-6, 11.967 h) vs cos θ (dim-5, 23.93 h). Bottom centre — density hierarchy: ρ_crit = ρ_max = αK·ρ_Pl (sub-Planckian, corrected). Bottom right — conditional consequences.

4. Decisive Laboratory Test

4.1 Interaction in the non-relativistic limit

$$H_{int} \sim \alpha K (\hat{u} \cdot I_{nuclear})^2$$

$\hat{u}$: fixed torsion axis in galactic frame | quadratic → modulation at $2\omega_{sid}$

4.2 Derivation of the 11.967 h period

As Earth rotates, $\theta(t) = \theta_0 + \omega_{sid} t$. The magnetic structure factor gives:

$$I(t) \propto I_0 [1 + \alpha K \cos^2\theta(t)]$$

$$\cos^2\theta = \tfrac{1}{2}(1 + \cos 2\theta) \rightarrow \text{time-varying part oscillates at } 2\omega_{sid}$$

Modulation period = $T_{sid}/2$ = 11.967 h | amplitude coefficient = $\alpha K/2$ = 85 ppm

$$\text{Peak-to-peak } (\Delta I/I) = 2 \times \alpha K/2 = \alpha K = 170 \text{ ppm}$$

Standard experimental convention — Series L

DISCRIMINANT: A dim-5 (linear) operator gives modulation at ω_{sid}, period 23.934 h. A dim-6 (quadratic) operator gives modulation at $2\omega_{sid}$, period 11.967 h. Observing the wrong period falsifies the dim-6 interpretation independently of the amplitude. This is a structural prediction, not data-fitting.

4.3 Experimental specifications

- Beamline: SPring-8 BL09XU, 7 keV, flux 10^{12} ph/s
- Sample: ^{57}Fe-doped diamond (nanodiamond or bulk)
- Reference: identical cell without ^{57}Fe (common-mode cancellation)
- Rotation: air-bearing stage, 2 revolutions per sidereal day (at $2\omega_{sid}$)
- Detection: dual-cell differential + software lock-in at $2\omega_{sid}$
- Sidereal discriminant: phase slips 3.93 min/day vs solar frame — no known systematic mimics this

$$\text{SNR} = 170 \text{ ppm} / \sqrt{50^2 + 8^2} \text{ ppm} = 3.4 \text{ in 720 h (30 days)}$$

SNR = 5.8 in 2160 h (90 days, 5σ)

KILL CONDITION: A null result with upper limit < 300 ppm at $2\omega_{sid}$ after 720 h falsifies $\alpha K = 0.00017$. The programme is dead. Publish the constraint.

5. Conditional Consequences and Cosmological Context

5.1 Conditional structure (the honest form)

The following statements are NOT evidence for αK. They are consequences valid if and only if the RDAC experiment returns a positive detection at 170 ± 50 ppm with the predicted 11.967 h period.

- Black hole cores: every BH interior contains r_core = ■_Pl/√αK = 1.24e-33 m.
- Prethermal plateaus: isolated quantum systems stall at E_total/2 via equipartition with a flat-DOS hidden sector (Series M derivation).
- Cosmological anomalies become follow-up predictions — not before.

5.2 Cosmological coincidences — context only

Several astrophysical residuals are of order 10■■ — the same scale as αK: DESI BAO offset (~0.6% ~ 35αK), CMB birefringence (~0.34°), PTA feature at ~18.5 nHz, GW echoes at Δt/t ~ 10■■. These coincidences motivated the hypothesis but do not test it. They will become follow-up predictions if RDAC returns positive. Until then: context only.

5.3 Inward-bang cosmology — one sentence

If the torsion condensate regularises every black hole interior, and if black holes can seed new universe-like regions via the Einstein-Cartan bounce (Poplawski 2010 [1]), then αK may set the scale of such regions — a speculation that becomes testable only after the RDAC experiment confirms αK as a real physical coupling.

6. Conclusions

We have presented a minimal extension of Einstein-Cartan gravity: a dynamical torsion condensate with a single dimensionless coupling αK = 0.00017. The condensate caps curvature at R_max ~ αK/■_Pl² and replaces the black hole singularity with a regular de Sitter core of radius r_core = ■_Pl/√αK = 1.239e-33 m. Crucially, the same αK predicts a sidereal modulation in ■■Fe-doped diamond with period 11.967 h (derived from the quadratic operator structure, not fitted) and peak-to-peak amplitude 170 ppm. A null result below 300 ppm in 720 h at SPring-8 BL09XU kills the hypothesis. A positive result opens a new window into the torsional vacuum. The data will decide.

SERIES N v2.2 — FINAL STATE

One operator: L = (αK/Λ²)(∂θ)²(ψ■γ∂ψ) — dim-6, CPT-even, shift-symmetric

One number: αK = 0.00017 — fit parameter until RDAC

One test: 170 ppm at 11.967 h — SPring-8, 720 h, kill < 300 ppm

Two conditional consequences (only if RDAC positive):

BH cores at r_core = 1.24e-33 m · Prethermal plateau at E/2

Cosmological coincidences: context only — not evidence.

Inward-bang cosmology: one conditional sentence citing Poplawski.

THE DATA WILL DECIDE.

Acknowledgments. The author thanks the MSDS programme internal audit (Omega Collective) for correcting earlier inconsistencies: the BAO factor-of-10 error (3.5αK → 35αK), the diamond signal amplitude (85 ppm → 170 ppm as peak-to-peak), the modulation period (23.934 h → 11.967 h, with derivation added), the trans-Planckian density inconsistency (ρ_crit = ρ_Pl/αK → αK·ρ_Pl), and the r_core numerical value (7.7×10■³³ m → 1.24×10■³³ m per the formula). This work is dedicated to the principle that one decisive experiment is worth a hundred phenomenological coincidences.

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